

TrES-3b

R-1544

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_190406 · created 2026-08-01T19:25:24+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458578.6178 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.193 +/- 0.057
Transit depth (Rp/Rs)^2	3.71 +/- 2.18 [%]
Orbital Inclination (inc)	83.44 +/- 2.82
Airmass coefficient 1 (a1)	1.14 +/- 0.13
Airmass coefficient 2 (a2)	-0.1 +/- 0.22
Scatter in the residuals of the lightcurve fit is	10.96 %
Best Comparison Star	#3 - [255, 444]
Optimal Aperture	6.11
Optimal Annulus	10.69
Transit Duration (day)	0.0209 +/- 0.0097

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.193 $\pm$ 0.057 vs 0.16309 (deviation 0.52 σ)
Duration fit vs published	0.0209 d vs 0.0567 d (deviation 3.69 σ)
Mid-time O-C	-1878.5 min
Depth SNR	3.4
Residual scatter	10.96 %

Light curve

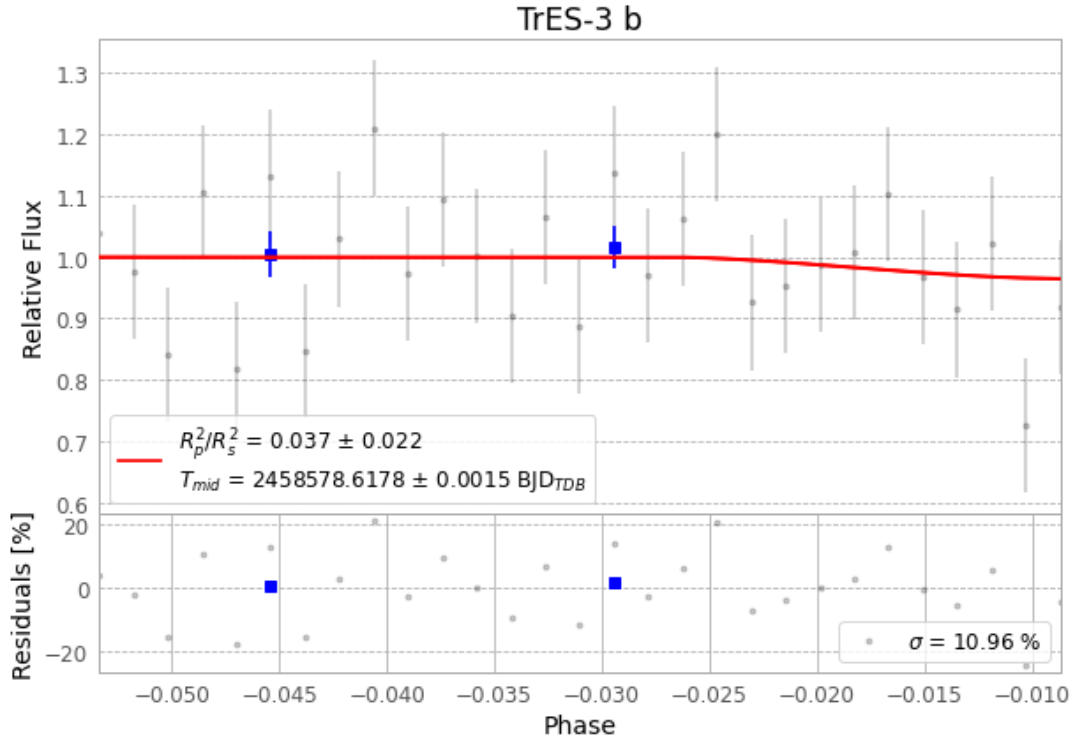


Figure 1 — FinalLightCurve_TrES-3 b_2019-04-06.png (SHA-256 d8f46af190290449...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [428, 103], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2ce6bf5d9baea481...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

66 FITS frames (43 MB), TRES-3190406082712.FITS → TRES-3190406114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-190406.log (df0411a30237...), FinalLightCurve_TrES-3 b_2019-04-06.png (d8f46af19029...), FinalLightCurve_TrES-3 b_2019-04-06.csv (ff5fb215a611...)

Compute resources

Wall time 220.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 19:25Z.